



Major cyber attack

Head of UK's National Cybersecurity Centre has warned that a major cyber attack will happen soon.
<http://dgit.in/cybrattck>



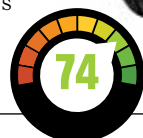
Autonomous dental implant

The world's first successful autonomous dental implant was carried out on woman in China.
<http://dgit.in/RoboSurgry>

Linksys EA7500 MAX-STREAM AC1900 Wi-Fi Router

Setting the benchmark for build quality

Linksys is well-known in the networking industry for their extremely well-built devices. There isn't a networking guy who wouldn't swear by the infamous WRT54G series. That thing was built like a rock and had a reputation for consistent performance through the passage of time. However, Linksys is also known for their ridiculous pricing, especially in India. So when the Linksys EA7500 came to us for a price tag of ₹13,499 (Amazon), we were a bit taken aback. Which is why we decided to take it apart to figure out the reason for the peculiar pricing.



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of a MediaTek IPQ8064 Dual-core processor which hosts a 880 MHz MIPS CPU, a 5-port Ethernet switch and handles the other two SoCs which control the individual bands.

The good thing about the MT7621AT is that it supports OpenWrt firmware.

Needless to say, we're quite impressed with the overall design of the inside. However, there are instances where the filter caps are electrolytic. We honestly feel that having a couple of tantalum caps in parallel would have been much better here since we face a lot of electricity spikes in India. And having electrolytic caps just begs to be the first point of failure. Then again, if you have access to a soldering iron, replacing these electrolytic caps are way easier than replacing tantalum SMD caps.

It turns out that the European and US SKUs of the Linksys EA7500 are built around the Qualcomm (Atheros) IPQ8064 as the main SoC along with Winbond NAND memory. The WLAN chips are again, Qualcomm QCA9982 and QCA9983. To put things into perspective, you can think of budget smartphones. Rarely do they come with Qualcomm Snapdragon SoCs and instead pack MediaTek SoCs which are way cheaper and "get the job done". In a similar fashion, even the EA7500

makes use of the more economical MediaTek MT7621AT.

Performance

Let's start with the USB ports for the storage. We were able to obtain about 68 MBps on the USB 3.0 port for read operations and 89 MBps for write operations. For routing via WLAN-LAN, we managed to get an average of 693 Mbps and near about the same while performing LAN-LAN transfer.

We then moved onto the individual bands. On the 2.4 GHz band, we managed to get an averaged uplink of 79 Mbps and around 83 Mbps on the downlink. While on the 5 GHz band, we managed to get 285 Mbps on the downlink and 195 Mbps on the uplink.

The UI is quite snappy and hardly falters. However, the lack of URL filtering in the web UI was a bit weird for a router this expensive. It's certainly not hard to get past URL filters in this day and age where all the major sites use HTTPS, but it certainly helps add a first layer of URL blocking for parents.

Verdict

Despite the MediaTek SoC, the Linksys EA7500 is still one of the best performing AC1900 routers we've tested, so in no way should you discount Linksys's capabilities to pull off a good design. From the competition, you have the NETGEAR R7000, the Tenda AC15, the ASUS AC68U and the TP-LINK C9. We'd say the NETGEAR R7000 is a pretty close competitor in terms of build and the ASUS RT-AC68U performs equally well or even better in certain benchmarks, however, they're all equally priced.

—Mithun Mohandas

SPECIFICATIONS

SOC: Mediatek MT7621AT | WLAN1: Mediatek MT7615N | WLAN2: Mediatek MT7615N | LAN: 4x Gigabit Ethernet | WAN: 1x Gigabit Ethernet | PROTOCOLS: 802.11ac (1900 Mbps).

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Packaging and Build Quality

The retail unit of the Linksys EA7500 comes with three detachable antennae, software disk, instruction manual, an Ethernet cable and the power adapter.

There's ample amount of cooling and these high gain routers definitely need all the cooling that they can get. The design of the router is very classy which has apparently become a huge factor for folks getting routers these days. Moreover, the plastic used for the body is quite thick and durable.

The rear of the router reveals that we have both, a USB 2.0 and a USB 3.0 port. Then there are the antennae connectors which are all gold plated. And lastly, the WPS button is quite easily accessed while the reset button is a bit recessed to avoid accidental resets. We have no complaints here.

Once opened, we could see why Linksys is known for their build quality. To start off, the PCB is by no means thin. And each band along with the main SoC is sectioned off with ample space between them to reduce interference. The Linksys EA7500 makes use